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# UPPER PALAEOZOIC PALAEOMAGNETIC RESULTS FROM ALGERIA AND TUNISIA

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## ABSTRACT

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Igneous and sedimentary rocks of Devonian, Carboniferous and Permian age from Algeria and Tunisia show evidence for remagnetisations that are partially removed or reduced by thermal demagnetisation, although alternating magnetic fields were ineffective in removing this effect from igneous rocks. Nonetheless, the remaining Devonian data are too sparse for adequate interpretation. Carboniferous sediments contain a magnetic component that can be isolated successfully and which appears to define a genuine Mid to Late Carboniferous shallow, southeasterly geomagnetic field direction. Although demagnetisation of Carboniferous igneous rocks was less successful, the same component appeared to be present in these rocks. Permian and Permo-Triassic sediments show only poorly defined, high blocking temperature directions, and those of Permian age do not appear to be magnetically consistent with other African data. It is thought that some of the directions isolated may be indicators of local tectonic rotations, but that it is premature to use these observations to define such rotations in the North African region.

## INTRODUCTION

The location of northern Africa during the Phanerozoic is of importance for understanding the evolution and development of much of Western Europe and the Mediterranean. In particular, its location relative to the European (Baltic) Shield is critical during the Hercynian orogeny and any understanding of the later development of the Western Mediterranean involves the palaeogeographic location of this region. Despite this significance, there are few palaeomagnetic data from this region, with the notable exception of those by Hailwood (1968, 1972), Daly and Pozzi (1976) and Martin et al. (1978). On this basis rock samples of a range of ages were collected from Algeria and Tunisia in 1973 and 1974 for palaeomagnetic analysis (Abou-Deeb, 1976), including some of Late Palaeozoic age from south of the South

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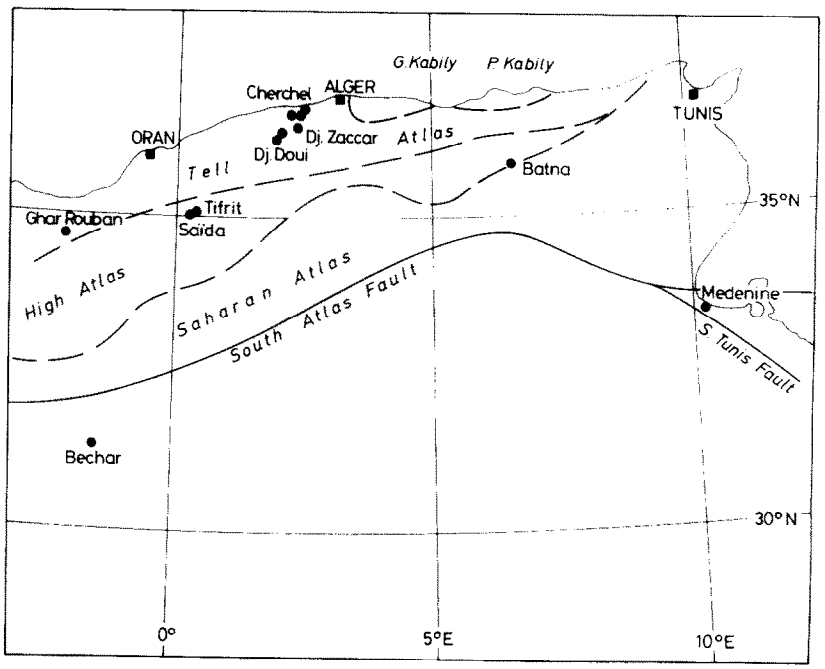



Fig. 1. Map of sampling locations. The dots refer to sampling locations, with the name used in the text for the nearest town, mountain, etc. Only the major structural elements are shown.

Atlas fault (Fig. 1). The methods of collection, measurement and analysis were standard (Collinson, 1983; Tarling, 1983). Cores were drilled at most sites, although hand samples were sometimes collected. Orientation was made using a sun compass whenever possible, although occasionally a magnetic compass had to be used. All the cores were later sliced to provide cylinders 2.54 cm diameter and 2.48 cm high. The remanence of these specimens was measured using a Digico spinner fluxgate magnetometer (Molyneux, 1971) with a noise-level of 0.02 $\mu$A/m. On this basis, measurements of standard specimens with an intensity of less than 0.2 $\mu$A/m have been ignored in the subsequent analysis. Partial demagnetisation of all sediments was undertaken thermally while igneous rocks were initially partially demagnetised using alternating magnetic field techniques and subsequently using thermal methods. During thermal demagnetisation, the low field susceptibilities were routinely monitored, using a bridge system (Collinson, 1983), as a first order indication of chemical changes affecting the magnetic minerals. The changes in the remanent vector during partial demagnetisation were analysed using computer routines to establish the components remaining or removed at each step and to estimate their stability, direction and linearity.

In this report, the results obtained from the analyses of Late Palaeozoic rocks (Fig. 1) are reported. A common feature of all of the alternating magnetic field

partial demagnetisation was that the majority of the igneous specimens examined showed little change from their initial direction, which was generally close to the present Earth's magnetic field direction in North Africa, even in the peak field exceeding 100 mT, but specimens cut from the same cores and subjected to thermal demagnetisation showed significant changes and vector analyses isolated different components, some of which appear to correspond with the original direction acquired as the igneous rocks cooled. This difference in behaviour to different demagnetisation methods is interpreted to mean that a secondary component was acquired during the last few million years and that it is stable to alternating magnetic fields, but less stable to thermal demagnetisation. On this basis, most of the discussion is concerned with the thermal demagnetisation behaviour.

DEVONIAN SEDIMENTS

Seven sites were sampled in Algeria, two (A33, A34) of black limestones exposed at the coast at Chenoua (36.6°N, 2.3°E) and five (A102–106) of folded sandy limestones at Ghar Rouban (34.6°N, 2.2°W), described by Lucas (1937). The intensity of magnetisation was generally weak, less than 0.1 $\mu$A/m in site A33, but mostly above 2 $\mu$A/m in site A34, with other sites ranging between these extremes. The mean directions (Table I) were poorly defined, with $\alpha_{95}$ greater than 30°, except for sites A34 and A106, which were quite well grouped. One pilot specimen was selected from sites A34 and A103 and thermally demagnetised. Both showed evidence for stability (Table II, Fig. 2a) and susceptibility changes above 600°C and 400°C respectively. Thermomagnetic analyses of crushed materials from these cores suggested that haematite was the dominant magnetic mineral, although irreversible inflexions at 270°C indicated that a low Curie temperature mineral was also present in A103. All of the Devonian samples were then heated to 200°C. This resulted in a considerable increase of the within-site scatter in site A33 and some increase in that of site A106, but an improvement of the scatter in most other sites (Table I, Fig. 3a). Nonetheless the circular standard deviation was greater than 20° in sites A33, A102 and A104, and it is not considered that these directions can be interpreted meaningfully. The site mean directions, after excluding three strongly deviant specimen directions, were almost exactly reversed in declination and of shallow positive inclination for sites A34 and A103, while the site mean direction for sites A105 and A106, after excluding two deviant specimen directions, were easterly and of shallow to steep positive inclinations. In view of this wide dispersion of the site mean directions, it is not thought that any useful interpretation can be made of these data at this stage as it is not clear to what extent the scatter is due to the continued presence of secondary components or to tectonic disturbances.

CARBONIFEROUS SEDIMENTS

In Algeria, south of the South Atlas fault, this period is represented by a very thick sedimentary sequence, comprising mostly limestones, sandstones and marls

TABLE I

Site mean directions of magnetisation *

| Site | $N$ | Decl. (°) | Incl. (°) | $\alpha_{95}$ (°) | $N$ | Decl. (°) | Incl. (°) | $\alpha_{95}$ (°) |
|---|---|---|---|---|---|---|---|---|
| *Devonian (sediments)* | | | | | | | | |
| A33 | 0 | – | – | – | 0 | – | – | – |
| A34 | 8 | 0.0 | 12.7 | 4.6 | 8 | 1.9 | 15.4 | 5.8 |
| A102 | 6 | N | +L | 90 | 3 | S | +L | – |
| A103 | 6 | SW | +I | 58 | 6 | S | +L | 19 |
| A104 | 2 | NW | +I | – | 0 | – | – | – |
| A105 | 4 | N | +I | 46 | 0 | – | – | – |
| A106 | 7 | 15.7 | 47.2 | 7.2 | 1 | N | +I | – |
| *Carboniferous (sediments)* | | | | | | | | |
| *Black limestones* | | | | | | | | |
| A84 | 6 | 28 | 68 | 11 | 6 | 106.2 | 68.5 | 7.4 |
| A85 | 6 | 138.8 | 39.4 | 9.8 | 6 | 139.6 | 34.6 | 8.6 |
| A87 | 6 | 120.4 | 13.1 | 7.4 | 6 | 126.2 | 11.8 | 8.6 |
| A88 | 6 | 126.5 | 11.0 | 6.8 | 6 | 131.3 | 8.3 | 6.0 |
| *Westphalian* | | | | | | | | |
| A92 | 6 | 29 | 52 | 12 | 5 | 135.0 | 17.3 | 7.5 |
| A93 | 6 | 352.7 | 15.7 | 2.2 | 1 | N | +L | – |
| *Namurian* | | | | | | | | |
| A94 | 6 | 3.3 | 47.1 | 7.5 | 1 | E | +I | – |
| A95 | 6 | 76.0 | 22.5 | 6.4 | 6 | 93.1 | 8.9 | 5.5 |
| A96 | 7 | E | +I | 19 | 7 | 118.5 | 10.8 | 3.7 |
| A97 | 6 | N | +L | 21 | 6 | N | +L | 18 |
| A98 | 6 | 12.3 | 35.9 | 8.0 | 0 | – | – | – |
| A99 | 5 | 6.6 | 25.7 | 5.3 | 0 | – | – | – |
| A100 | 5 | 340.5 | 18.4 | 2.9 | 5 | 244 | 11 | 14 |
| A101 | 5 | NE | +I | 27 | 1 | SE | +L | – |
| *Carboniferous (igneous)* | | | | | | | | |
| *Rhyolite* | | | | | | | | |
| A35 | 6 | E | +S | 25 | 6 | E | +S | 35 |
| A70 | 6 | 8.7 | 53.0 | 5.1 | 6 | 293.3 | 37.8 | 7.9 |
| A71 | 7 | 29 | 48 | 14 | 7 | N | +L | 62 |
| A86 | 5 | 121.6 | 19.8 | 5.2 | 5 | 131 | 21 | 11 |
| A89 | 6 | 331.0 | 49.1 | 7.5 | 6 | S | +I | 32 |
| A90 | 5 | E | +I | 25 | 4 | 157 | 44 | 16 |
| *Andesites* | | | | | | | | |
| A40 | 6 | 343.0 | 56.8 | 6.5 | 6 | 259 | 36 | 19 |
| A41 | 6 | N | +I | 25 | 6 | 100.5 | −13.6 | 9.7 |
| A42 | 6 | N | +S | 21 | 6 | SE | +I | 21 |
| A43 | 5 | −20.7 | 54.1 | 8.0 | 5 | 132 | 39 | 17 |
| A44 | 6 | 356 | 53 | 11 | 6 | 15.6 | 35.3 | 6.1 |

TABLE I (continued)

| Site | $N$ | Decl. (°) | Incl. (°) | $\alpha_{95}$ (°) | $N$ | Decl. (°) | Incl. (°) | $\alpha_{95}$ (°) |
|---|---|---|---|---|---|---|---|---|
| A45 | 7 | 19.5 | 41.5 | 6.7 | 7 | 31 | 26 | 12 |
| A47 | 6 | 41 | 14 | 13 | 6 | 40 | 11 | 14 |
| A48 | 4 | N * | +L | 31 | 2 | NE | −L | – |
| A51 | 6 | 22 | 48 | 12 | 6 | 24 | 35 | 19 |
| A56 | 6 | N | +I | 18 | 1 | NE | −L | – |
| A57 | 6 | 14.2 | 36.3 | 9.0 | 5 | 35 | 29 | 15 |
| *Diorite* | | | | | | | | |
| A36 | 6 | E | +I | 52 | 6 | NE | +L | 32 |
| A37 | 6 | 223.8 | 67.6 | 7.9 | 6 | 228 | 57 | 11 |
| *Granodiorite* | | | | | | | | |
| A75 | 6 | N | +S | 23 | 6 | SE | +I | 21 |
| A76 | 6 | E | +I | 30 | 6 | W | −L | 41 |
| *Breccia* | | | | | | | | |
| A82 | 6 | 139.8 | 27.8 | 4.5 | 6 | 142.6 | 23.5 | 2.1 |
| A38 | 6 | 138.8 | 32.1 | 6.5 | 6 | 153.3 | 20.8 | 8.2 |
| *Granite* | | | | | | | | |
| A50 | 6 | S | +I | 90 | 0 | – | – | – |
| A61 | 6 | 82.8 | 50.1 | 7.4 | 6 | 114.2 | 53.3 | 8.9 |
| A62 | 6 | 260 | 41 | 15 | 6 | 145 | 51 | 11 |
| A63 | 6 | N | +L | 20 | 6 | 37 | 72 | 16 |
| A65 | 5 | SE | +L | 33 | 5 | S | +I | 24 |
| A66 | 6 | 148 | 28 | 16 | 6 | SE | +L | 39 |
| *Permian (sediments)* | | | | | | | | |
| *Red Sandstones* | | | | | | | | |
| T37 | 6 | 354 | −26 | 13 | 0 | – | – | – |
| T38 | 6 | 3.5 | −23.7 | 1.4 | 4 | 9 | −16 | 11 |
| T40 | 6 | 1.5 | −7.7 | 8.0 | 0 | – | – | – |
| T41 | 6 | 354.2 | 50.9 | 3.8 | 6 | 12.2 | −16.4 | 8.5 |
| T43 | 6 | N | −L | 23 | 6 | 338 | −12 | 14 |
| T44 | 6 | 335 | 25 | 16 | 2 | – | – | – |
| *Carbonates* | | | | | | | | |
| T39 | 6 | 350.3 | −23.7 | 9.2 | 0 | – | – | – |
| T42 | 6 | 345 | 13 | 12 | 0 | – | – | – |
| *Schists* | | | | | | | | |
| T35 | 6 | 342.0 | −6.6 | 4.3 | 6 | 336.9 | −7.2 | 6.6 |
| T36 | 6 | 343.8 | 1.2 | 2.3 | 1 | SW | +L | – |
| *Permo-Triassic (sediments)* | | | | | | | | |
| *Red Sandstone* | | | | | | | | |
| A30 | 6 | 336.3 | 3.9 | 8.6 | 6 | N | −I | 24 |
| A31 | 2 | 11.7 | 0.9 | – | 0 | – | – | – |
| A32 | 6 | 55.5 | −14.7 | 7.8 | 6 | 58.4 | −17.7 | 8.4 |

TABLE I (continued)

| Site | $N$ | Decl. (°) | Incl. (°) | $\alpha_{95}$ (°) | $N$ | Decl. (°) | Incl. (°) | $\alpha_{95}$ (°) |
|---|---|---|---|---|---|---|---|---|
| A39 | 6 | 28.9 | 10.5 | 4.6 | 6 | 32.8 | 9.8 | 5.7 |
| A52 | 6 | 16.0 | 18.1 | 3.0 | 4 | 38 | 9 | 12 |
| A53 | 6 | NW | +I | 70 | 6 | SW | +I | 87 |
| A54 | 5 | W | +L | 36 | 5 | W | +L | 45 |
| *Carbonates* | | | | | | | | |
| A8 | 6 | 58 | −9 | 17 | 6 | 105 | −15 | 12 |
| A9 | 6 | N | −L | 25 | 4 | 303 | −55 | 14 |

* Mean directions have been calculated after rejection of specimens with an intensity of less than 0.2 $\mu$A/m. The number of samples per site is $N$. The corresponding mean directions are only approximated if the circular standard deviation is greater than 20°, and the direction is to the nearest degree if the $\alpha_{95}$ value is greater than 10°. Directions and $\alpha_{95}$ values are not given for N less than 3 unless very well grouped. L = low, I = intermediate, S = steep.

(Menchikoff, 1949). Two sites were sampled in Lower Westphalian limestones and eight sites in Namurian blue limestones near Colomb Bechar (31.3°N, 1.5°W). Black sandy limestones were also sampled at four sites in a thick Carboniferous series in the Tifrit Complex (Lucas, 1952) near Saida (35.0°N, 0.3°E).

The initial intensities of remanence of these limestone and sandy limestone samples were generally low, all less than 20 $\mu$A/m, and two sites were characterised by intensities below the instrumental level and three other site means were hardly definable (Table I). The remaining site mean directions were quite widely scattered despite several sites having low internal scatters. Four specimens were subjected to partial thermal demagnetisation (Table II, Fig. 2b), with each showing somewhat different behaviour. Specimen A97.2 showed only moderate stability, with directions loosely grouped near, but somewhat shallower than the local geomagnetic field. Specimens A84.1 and A93.1 both showed stability, but A84.1 had directions close to the ambient geomagnetic field until chemical changes, indicated by increases in its susceptibility, occurred and were associated with an increase in directional scatter. A93.1 had directions which moved systematically towards a shallow southeast direction (160°, 8°), but above 500°C the susceptibility began to increase and the direction change became erratic. High stability characterised specimen A87.1, with little change in either direction or intensity until temperatures in excess of 400°C, and no significant changes until over 550°C.

The behaviour of the pilot specimens, particularly the most stably magnetised specimens A93.1 and A87.1, indicates a common component in a shallow southeasterly direction. Thermal demagnetisation of all the specimens at 200°C showed that

TABLE II

Stability characteristics of pilot specimens *

| Sample | Decl. (°) | Incl. (°) | Range (°C) | Max. (1) S.I. | Range (°C) | Max. (2) B.I. | No. linear segments |
|---|---|---|---|---|---|---|---|
| *Devonian sediments* | | | | | | | |
| A34.9 | 1.1 | 17.9 | 50–200 | 3.3 | 50–100 | 0.94 | 2 |
| A103.1 | 187.4 | 14.3 | 20–300 | 15.4 | 50–100 | 0.96 | 1 |
| *Carboniferous sediments* | | | | | | | |
| A84.1 | 99.2 | 69.7 | 200–400 | 1.6 | 50–100 | 0.86 | 3 |
| A87.1 | 121.3 | 17.6 | 100–300 | 6.0 | 100–200 | 0.96 | 1 |
| A93.1 | 8.2 | 39.4 | 100–500 | 1.3 | 100–200 | 0.59 | 3 |
| A97.2 | 337.1 | 7.0 | 200–400 | 1.0 | 200–300 | 0.63 | 2 |
| *Carboniferous, igneous* | | | | | | | |
| A35.1 | 359.9 | 57.7 | 300–600 | 5.1 | 300–400 | 0.89 | 2 |
| A71.7 | 9.9 | 38.0 | 20–300 | 2.8 | 200–300 | 0.74 | 2 |
| A40.1 | 295.2 | 81.5 | 20–300 | 1.5 | 400–500 | 0.78 | 1 |
| A45.1 | 30.2 | 20.0 | 200–500 | 1.6 | 200–300 | 0.73 | 0 |
| A47.2 | 43.3 | 9.6 | 100–500 | 10.7 | 400–500 | 0.97 | 3 |
| A37.1 | 238.5 | 64.7 | 100–300 | 3.6 | 400–500 | 0.86 | 2 |
| A82.1 | 143.0 | 22.1 | 100–300 | 4.3 | 100–200 | 0.96 | 3 |
| A61.1 | 107.6 | 48.5 | 200–400 | 7.0 | 200–300 | 0.93 | 1 |
| *Permian* | | | | | | | |
| T41.3 | 25.6 | −7.8 | 100–550 | 7.5 | 300–400 | 0.93 | 1 |
| T44.1 | 157.6 | −25.1 | 100–400 | 1.6 | 300–400 | 0.80 | 1 |
| T36.2 | 335.4 | 10.4 | 20–100 | 0.5 | 50–100 | 0.03 | 1 |
| *Permo-Triassic sediments* | | | | | | | |
| A8.5 | 31.4 | −3.0 | 20–100 | 0.9 | 50–100 | 0.88 | 2 |
| A39.3 | 31.3 | 18.3 | 200–600 | 14.1 | 500–550 | 0.96 | 2 |
| A52.1 | 27.6 | 15.8 | 100–300 | 3.4 | 100–200 | 0.51 | 3 |
| A32.2 | 65.8 | −13.1 | 100–400 | 11.0 | 50–100 | 0.87 | 1 |

* The most stable direction for each specimen is given based on (1) the maximum Stability Index (Tarling and Symons, 1967) and the temperature range (°C) over which this is defined. The temperature ranges for (2) the maximum Briden Index (Briden, 1972) are also tabulated. The number of non-overlapping linear segments are also given.

most sites had mean directions in this region (Fig. 3b), although several specimens had become too weak to measure. The five best grouped sites, A85, A87, A88, A92 and A96, had a mean direction of 129.7°, 16.6°, $\alpha_{95} = 12.5°$. Inclusion of the two nearer site mean directions, A84 and A95, changed the mean direction to 122.5°, 22.9°, and increased the circle of 95% confidence to 20.5°. It is considered that the evidence for the pilot specimen behaviour and the grouping of the site means

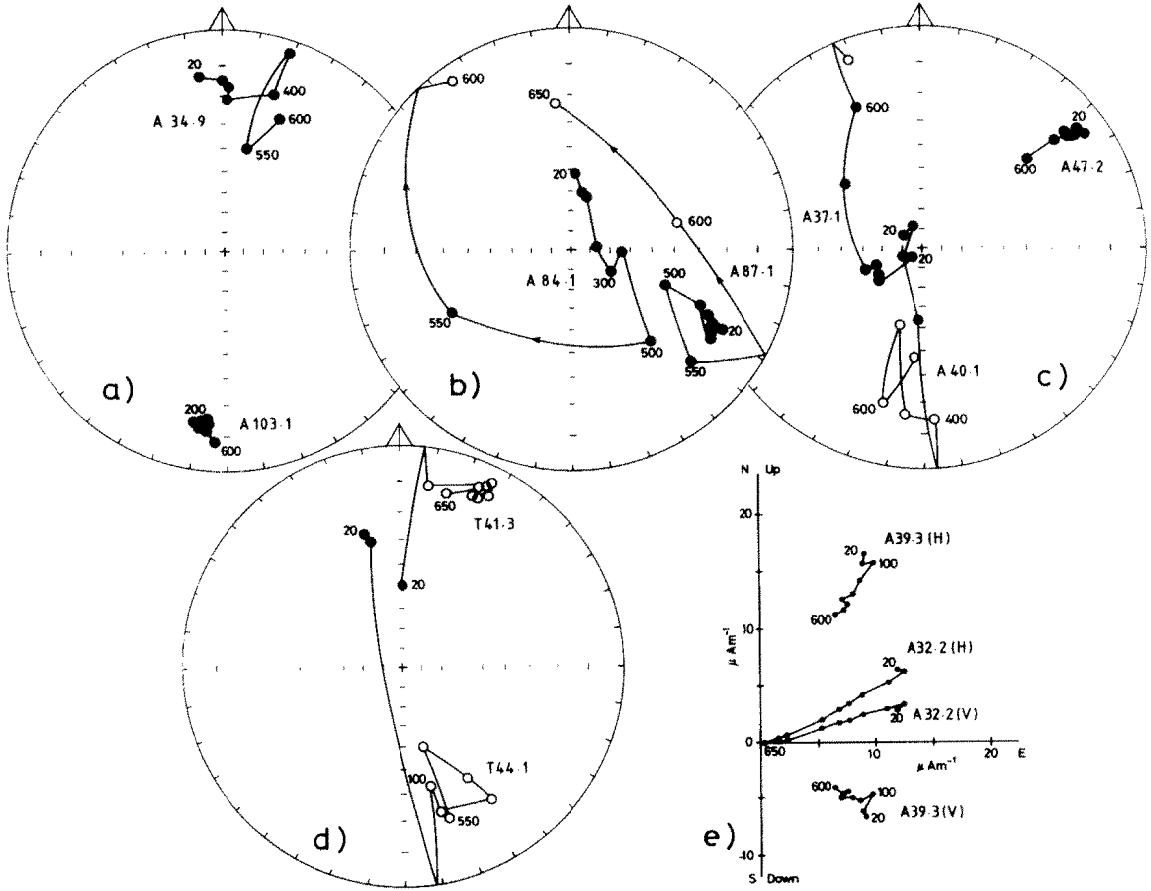



Fig. 2. Examples of pilot specimen behaviour. (a) Devonian, (b) Carboniferous sediments, (c) Carboniferous igneous rocks, (d) Permian, and (e) Permo-Triassic. The stereographic projections are equal angle, with solid symbols indicating positive inclinations. The Cartesian coordinate diagram (e) shows all the points as dotted. The temperatures are marked in °C.

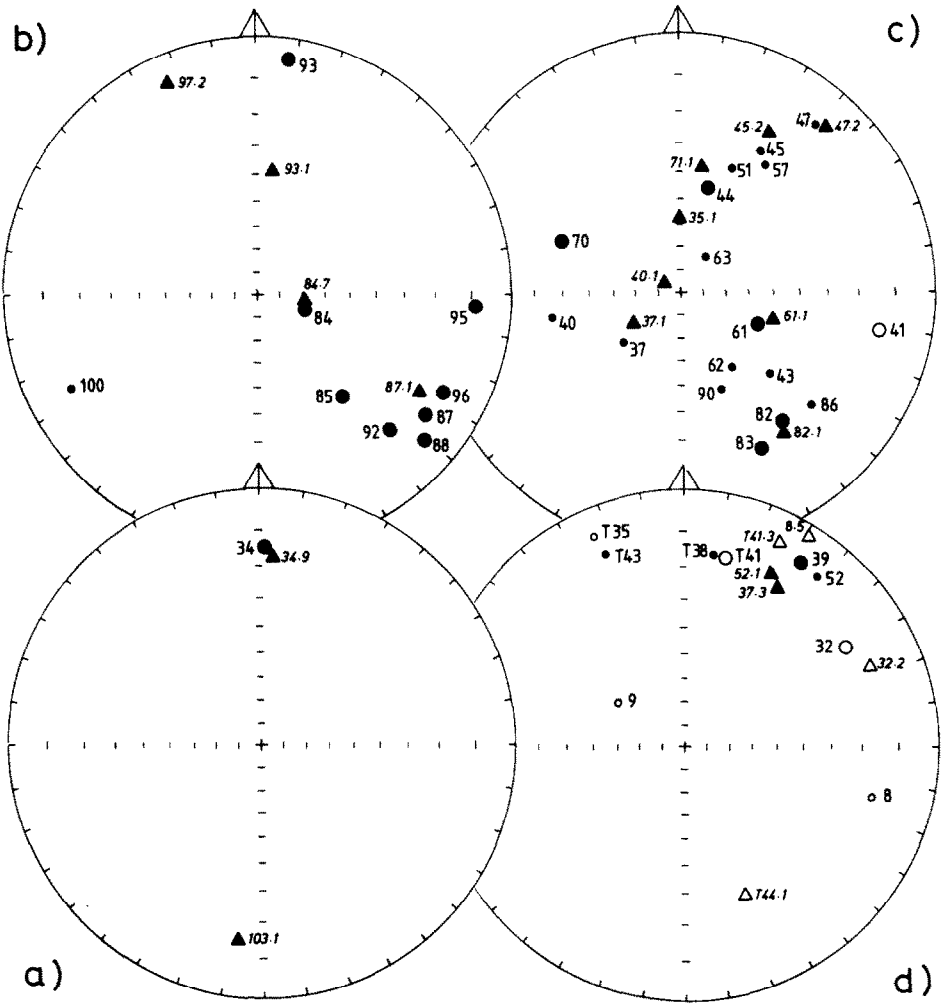



Fig. 3. Most stable and cleaned site mean directions. The samples illustrated are all from Algerian sites (A) unless specifically labelled as *T* (Tunisian). The net is an equal angle stereographic polar projection.

indicates that the geomagnetic field in Algeria during Namurian–Westphalian times was southeasterly and shallow; probably best represented by the five most closely grouped site mean directions.

## CARBONIFEROUS IGNEOUS ROCKS

Samples were obtained from four Algerian localities and ranged from granites to rhyolites, although most were andesitic lavas. Most samples came from 14 sites (A61–63, A65, A66, A70, A71, A75, A76, A82, A83, A86, A89 and A90) in mainly

Visean age rocks exposed in a sequence at Tifrit (35.0°N, 0.2°E). Three sites (A35–37) were of undifferentiated Carboniferous volcanics near Cherchel (36.6°N, 2.2°E). The remaining 12 sites were sampled in two localities but the sequences are thought to be related, comprising volcanics erupted in two phases, predominantly in the Visean—sites A40–45 at Djebel Zaccar (36.4°N, 2.3°E) and sites A47, A48, A50, A51, A56 and A57 at Djebel Doui (36.3°N, 2.0°E).

The intensities of remanence of all the igneous specimens showed a wide range although most were well above the noise level of the magnetometer. Their directions, before demagnetisation were mostly close to the present geomagnetic field direction in Algeria (Table I). Eight pilot specimens were selected, each from a different site, to evaluate their stability characteristics to partial demagnetisation in alternating magnetic fields. All samples showed good stability, but the directions remained close to their original directions, mostly near the present geomagnetic field in Algeria. In view of the apparent stability, all the remaining specimens were partially demagnetised in a peak alternating field of 22.5 mT., but this treatment did not drastically alter the mean directions of remanence which remained close to the present geomagnetic field direction in Algeria. Clearly such a direction is inconsistent with most considerations of the probable palaeolatitude and orientation of northern Africa at this time. Despite their apparent stability, therefore, it appeared that such alternating magnetic field demagnetisation had not isolated the original components of remanence, but were still dominated by components acquired recently.

A further eight cores were cut from cores from the same sites that had previously been examined using alternating magnetic fields, and these were then subjected to thermal demagnetisation. All of these specimens also showed some stability to thermal demagnetisation, but there were changes in the initial directions (Table II, Fig. 2c). The most stable components had directions which appear to string from a shallow northeasterly direction towards and slightly beyond the present geomagnetic field direction, with two other sites possibly strung from the present field directions towards the southeast. There was no clear distinction between the different rock types. As it was not possible to obtain fresh specimens from the remaining cores, all the specimens that had previously been partially demagnetised in an alternating magnetic field of 22.5 mT were then heated to 200°C and their remaining remanence measured. This caused the scatter within some sites to decrease and increase in others, but it was not clear that this was the optimum treatment. Heating all the specimens to 300°C and then to 350°C indicated little or no difference between the directions isolated at all three levels, and so the directions isolated at 200°C have been taken as representative of the stable directions within each site (Table I, Fig. 3c). These directions also mirrored the directions isolated during partial thermal demagnetisation of the pilot specimens.

The observed scatter of the stable site means and of the magnetisation isolated in the pilot specimens is thought to reflect the presence of several components, in particular a direction in the approximate direction of the present geomagnetic field

which does not appear to have been completely removed in several sites. Comparison with the directions isolated for the Carboniferous sediments would suggest that the stringing towards the southeast is due to the presence of genuine Carboniferous components, but it is not thought that it is justified to combine arbitrarily selected site mean directions to determine this direction. However, the close agreement of the mean directions for the volcanic breccia, and their stability, would suggest that these have most effectively retained the original Carboniferous magnetisation.

PERMIAN SEDIMENTS

Middle and Upper Permian rocks (Solignac and Berkaloff, 1934; Giry and Mathieu, 1947; Mathieu, 1949) were sampled at nine sites in Tunisia at 33.5°N, 10.2°E. These comprised two sites (T35, 36) in red schists in the core of an anticline at the foot of Djebel Oum el Afia, four sites (T37, 38, 40, 41) in red sandstones and one (T39) in dolomite at Baten beni Zid, and one limestone (T42) and two red sandstone sites (T43, 44) in the Chequimi valley. The intensity of remanence was low, but all specimen values fell within acceptable limits prior to demagnetisation. The directions were well grouped within most sites and were characterised by generally northerly, shallow negative inclinations (Table I). Three pilot specimens were selected, two from red sandstones and one from schists (Table II, Fig. 2d). The schist specimen lost any measurable remanence above 200°C and was unstable. One of the sandstone pilot specimens (T44.1) also became unmeasurable above 550°C, although a moderately stable direction had been isolated before then. The other sandstone specimen showed high stability. Thermomagnetic analyses of all three specimens indicated the presence of two magnetic components, with Curie points of about 400°C and 600°C., although the weak magnetisation prevented precise determinations.

Thermal demagnetisation at 200°C reduced the intensity of remanence below measurement levels in both limestone sites, one of the schist sites and three of the red sandstone sites (Table I, Fig. 3d). The remaining sites, three red sandstones and one schist, had broadly similar northerly, shallow negative directions. The mean direction, giving equal weight to each site, is 353.7°, −13.2°, $\alpha_{95} = 22°$, and appears to be representative of the Mid–Upper Permian geomagnetic field in this region of Tunisia.

PERMO-TRIASSIC SEDIMENTS

Black limestones were sampled in Algeria at two sites (A8, A9) on the Ain M'lila to Batna road (35.8°N, 6.5°E). Red sandstones were sampled at three localities; three sites (A30–32) at the coastal exposure of the Chenoua massif (36.7°N, 2.4°E), one site (A39) at Djebel Zaccar (36.4°N, 2.3°E), and three sites (A52–54) near Sidi Daoud in Djebel Doui (36.2°N, 1.9°E). The initial intensities of magnetisation were

low to moderate and the mean directions were virtually undefined at three sites (A31, A53 and A54), although the remaining site directions were generally northerly and shallow (Table I). Pilot specimens were selected from one limestone and three red sandstone sites (Table II, Fig. 2e). The limestone specimen became too weak for adequate measurement above 300°C and no stable component could be isolated. All three sandstones showed high stability and they were all grouped in a northeasterly direction with shallow, positive and negative, inclinations. These three also had thermomagnetic curves with haematite Curie points, but also a lower Curie temperature mineral, probably an ilmeno-haematite, with a Curie temperature near 400°C.

Thermal demagnetisation at 200°C somewhat reduced the internal scatter in the limestone sites (Table I, Fig. 3d), although the scatter was still large and the agreement between the two sites was not good. The internal scatter within the sandstone sites was generally increased by thermal demagnetisation and only three sites retained adequately defined directions, which were northeasterly with both negative and positive shallow inclinations. The mean direction of these three sites is 42.9°, 0.5°, $\alpha_{95} = 32°$, and is thought to be only broadly indicative of the probable direction of the Permo-Triassic field in Algeria.

## COMPARISON WITH OTHER DATA

As there are few reliable data from the rest of Africa (Table III) and some of the areas sampled lie in complex structural regions, it is not easy to determine the extent to which differences between these and previous data can be attributed to local tectonic rotations or to the inadequacy of either these new data or the previous data as indicating the past geomagnetic field. The Devonian data from Algeria is, in any case, too unreliable and differences with that of the Msissi Norite in Morocco (Hailwood, 1974) probably reflect remagnetisations, etc., rather than genuine tectonic rotations between the areas involved.

The Permian data from Tunisia are somewhat scattered, but their corresponding palaeomagnetic pole position differs strongly from that of other African Permian poles (Fig. 4) and the northerly declinations and shallow negative inclinations correspond to a normal magnetisation for this region as it would have been just south of the Equator at this time. The palaeolatitude (6.7°S) appears somewhat inconsistent with that of the red sandstones in Tunisia (25°S) and from that suggested by the Dwyka pole positions (39°S), although similar to that suggested by the Ecca beds (7°S). The presence of a normal polarity during this period is inconsistent with the magnetisation having been acquired during the Kiaman reversed period. All the evidence therefore suggests that these rocks have been remagnetised at some later time, probably in the Triassic or the Cretaceous normal polarity epoch. If this is correct, then clearly any interpretation of this direction in terms of a Permian reconstruction would be misleading. The Permo-Triassic results are somewhat scattered and the number of reliable sites is only three so that it would

TABLE III

African Upper Palaeozoic pole positions

| | | $N$ | $\alpha_{95}/A_{95}$ | Lat. (°) | Long. (°) | Reference |
|---|---|---|---|---|---|---|
| *Devonian* | | | | | | |
| Msissi Norite | Morocco | 7 | 12 | 1 | 205 | Hailwood (1974) |
| *Carboniferous* | | | | | | |
| Meseta, etc. | Morocco | 22 | 6 | 5 | 236 | Martin et al. (1978) |
| Ain-Ech-Chebbi | Sahara | 1 | – | 25 | 235 | Conrad and Westphal (1975) |
| Sediments + igneous | Algeria | 5+ | – | 28 | 238 | this paper |
| *Permian* | | | | | | |
| Ecca K3 | E. Africa | 1 | – | 42 | 241 | Nairn (1964) |
| Ecca K3 | E. Africa | 9 | 16 | 40 | 244 | Opdyke (1964) |
| Dwyka varves | C. Africa | 5 | 11 | 27 | 207 | McElhinny and Opdyke (1968) |
| Chougrane and others | Morocco | 57 | 12 | 32 | 242 | Daly and Pozzi (1976), Martin et al. (1978) |
| M. Perm. sediments | Tunisia | 4 | | 63 | 204 | this paper |
| *Permo-Triassic* | | | | | | |
| Cassanje | Angola | 2 | – | 53 | 253 | Valencio et al. (1978) |
| Sediments | Algeria | 3 | – | 36 | 125 | this paper |

* Only the data for stable sites of reasonably well defined ages are listed. $N$ corresponds to the number of sites involved and $\alpha_{95}$ and $A_{95}$ are the radii of 95% confidence about the mean direction or pole position where available.

be premature to interpret these observations in terms of the possible post-Triassic tectonic rotations of this part of Africa. This means that the only data that can be constructively interpreted at this stage is that for the Carboniferous for which the geomagnetic field direction appears to be quite well defined by some of the sediments and this direction is generally confirmed by the igneous rocks, despite their severe overprint magnetisations.

Carboniferous sedimentary sites have been sampled both north and south of the major South Atlas fault (Fig. 1), but only one of the sites from south of the fault is considered sufficiently reliable for further analysis. It is doubtful if much weight can be placed on a single site, although the mean direction is consistent with those from north of the fault. These five sites have a mean pole position (Table III) that is almost identical with that obtained from the Sahara (Conrad and Westphal, 1975), although differing from that for a variety of rock types in Morocco (Martin et al.,

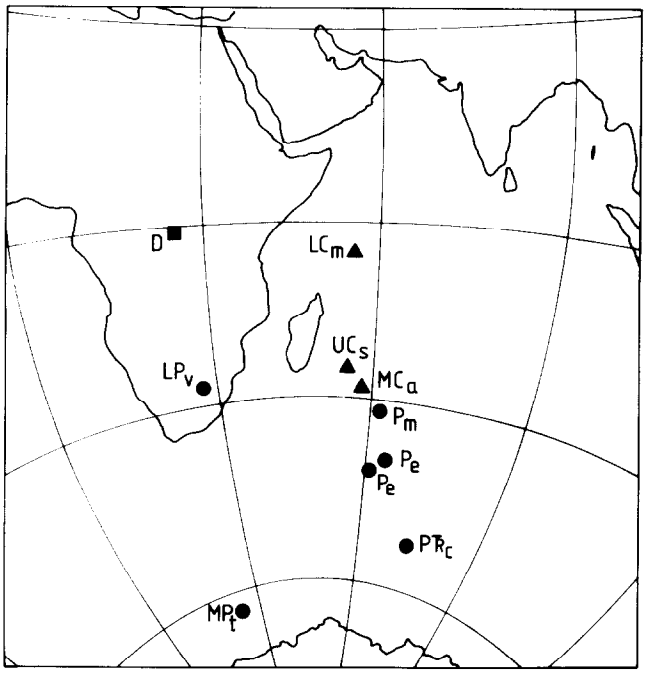



Fig. 4. Pole positions. The poles illustrated are listed in Table III. The square refers to Devonian, triangles to Carboniferous and dots are Permian or Permo-Triassic poles. The abbreviations, L, M and U refer to lower, middle and upper, and the subscripts refer to the initial letter of the formation.

1978). This difference could be interpreted in terms of an anticlockwise rotation of Morocco relative to the African block as postulated by Conrad and Westphal (1975), but it seems probable that the Rif and Atlas mountains represent a very complex sequence of dislocations within which small individual blocks are likely to have undergone a series of separate movements since the Palaeozoic in the same way as demonstrated for Lebanon (Freund and Tarling, 1979). It thus seems premature to draw megascale conclusions from palaeomagnetic data that must still be considered too poorly defined for meaningful tectonic interpretation although clearly such rotations appear to be highly probable in such a complex area and obviously require intensive as well as extensive further study.

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